

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

**COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I J. Kirithiga / 192421265 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. 19/20
I understand that any violation of academic integrity rules will result in disciplinary action.

J. Kirithiga
Signature of the student:

[Signature]

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on

the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem

CO2 - ASSESSMENT TOOL - 1

Given network:

Network Address: 192.168.20.0/24

Total IP Addresses = 256 (192.168.20.0 - 192.168.20.255)

The Network is divided using fixed length subnet mask (FLSM)

• LAB A: /26

• LAB B: /27

• LAB C: /28

LAB A (/26)

Subnet mask: 255.255.255.192

Block Size: 64

Network Address: 192.168.20.0

Valid Host range

first host: 192.168.20.1

Last host: 192.168.20.62

Broadcast Address:

192.168.20.63

Usable Host:

formula = $2^{(32-26)} - 2$

= $2^6 - 2 = 64 - 2 = 62$

Total Usable Host = 62

LAB B (/27)

Subnet mask: 255.255.255.224

Block Size: 32

The next available Subnet Start after Lab A
Network Address: 192.168.20.64

Valid host range:

first host: 192.168.20.65

Last host: 192.168.20.94

Broadcast Address:

192.168.20.95

Usable hosts: formula = $2^{(32-37)} - 2$
 $= 2^5 - 2 = 32 - 2 = 30.$

Total usable host = 30.

LAB C (128)

Subnet Mask: 255.255.255.040

Block Size: 16

The next available Subnet Start after LAB B.

Network Address: 192.168.20.96

Valid host range:

first host: 192.168.20.97

Last host: 192.168.20.110

Broadcast Address:

192.168.20.111

Usable hosts: formula = $2^{(32-28)} - 2$

$= 2^4 - 2 = 16 - 2 = 14$

Total usable host = 14

LAB A: Network = $192.168.20.0/26$, Host Range = $192.168.20.1 - 192.168.20.62$, Broadcast = $192.168.20.63$, 62 usable hosts
 LAB B: Network = $192.168.20.64/27$, Host Range = $192.168.20.65 - 192.168.20.94$, Broadcast = $192.168.20.95$, 30 usable hosts
 LAB C: Network = $192.168.20.96/28$, Host Range = $192.168.20.97 - 192.168.20.110$, Broadcast = $192.168.20.111$, 14 usable hosts.

2) Given:

System 1 IP Address: $192.168.1.130$

System 2 IP Address: $192.168.1.190$

System 3 IP Address:

Subnet Mask (both): $255.255.255.192$ (126)

Step 1: Calculate the Subnet size

Prefix: /26, Block Size = 64

Subnets are:

Subnet	Network Address	Host Range	Broadcast Address
1	$192.168.1.0$	$192.168.1.1 - 192.168.1.62$	$192.168.1.63$
2	$192.168.1.64$	$192.168.1.65 - 192.168.1.126$	$192.168.1.127$
3	$192.168.1.128$	$192.168.1.129 - 192.168.1.190$	$192.168.1.191$
4	$192.168.1.192$	$192.168.1.193 - 192.168.1.254$	$192.168.1.255$

System 1 : Analysis

IP Address : 192.168.1.130

Since 130 lies between 128 and 191, it belongs to

Network Address : 192.168.1.128

Subnet Mask : 255.255.255.192 (126)

Valid Host Range : 192.168.1.129 - 192.168.1.190

Broadcast Address : 192.168.1.191

Verification:

Given IP = 192.168.1.130

Host range = 192.168.1.129 - 192.168.1.190

Result : The IP address falls within its Subnet

System 2 : Analysis

IP Address : 192.168.1.190

Since 190 also lies between 128 and 191, it belongs to the

Same Subnet

Network Address : 192.168.1.128

Subnet Mask : 255.255.255.192 (126)

Valid Host Range : 192.168.1.129 - 192.168.1.190

Broadcast Address : 192.168.1.191

Verification:

Given IP = 192.168.1.190

Host Range = 192.168.1.129 - 192.168.1.190

The IP address falls within its Subnet

usable Host calculation
for a /26 Subnet

$$\text{Host bits} = 32 - 26 = 6$$

$$\text{Total Addresses} = 2^6 = 64$$

$$\text{Network Addresses} = 1$$

$$\text{Broadcast Address} = 1$$

$$\text{Usable Host} = 64 - 2 = 62$$

3) IP Subnetting Solution

Given:

Network block: 172.16.0.0/16

Departments and Subnet Masks

HR : /25

Sales : /26

IT : /27

Admin: /28

The Subnets are allocated sequentially from the beginning of the Network

1. HR Department (/25)

Subnet mask: 255.255.255.128

Network Address: 172.16.0.0

Valid Host Range: 172.16.0.1 - 172.16.0.126

Broadcast Address: 172.16.0.127

Usable hosts:

$$\text{formula} = 2^{(32-25)} - 2 = 2^7 - 2 = 128 - 2 = 126$$

Total Usable Host = 126

2. Sales Department (126)

Subnet Mask : 255.255.255.192

Network Address : 172.16.0.128

Valid Host Range : 172.16.0.129 - 172.16.0.190

Broadcast Address : 172.16.0.191

Usable Host :

$$\text{formula} = 2^{(32-26)} - 2$$

$$= 2^6 - 2 = 64 - 2 = 62$$

3. IT Department (127)

Subnet Mask : 255.255.255.224

Network Address : 172.16.0.192

Valid Host Range : 172.16.0.193 - 172.16.0.223

Broadcast Address : 172.16.0.223

Usable Hosts :

$$\text{formula} = 2^{(32-26)} - 2$$

$$= 2^6 - 2 = 32 - 2 = 30.$$

4. Admin Department (129)

Subnet Mask : 255.255.255.240

Network Address : 172.16.0.224

Valid Host Range : 172.16.0.225 - 172.16.0.238

Broadcast Address : 172.16.0.239

Total Usable Host = 14

$$= 2^4 - 2 = 16 - 2 = 14$$

$$\text{formula} = 2^{(32-26)} - 2$$

remaining unused IP Address Range
the allocated Subnets occupy addresses
 $172.16.0.0 - 172.16.0.239$

The remaining unused address within the $172.16.0.0/16$ network
are $172.16.0.240 - 172.16.255.255$

These addresses are available for future Subnet allocation

4) Given:

Network Address: $172.16.10.0/24$

Subnet Mask: $/27$ ($255.255.255.224$)

Step 1: Calculate the total number of Subnets

Original network = $/24$

New Subnet mask = $/27$

Borrowed bits = $27 - 24 = 3$

Total Subnets = 8

Step 2: Calculate the number of Usable Hosts per Subnet

Host bits = $32 - 27 = 5$

Total IP address per Subnet = $2^5 = 32$

Usable host = $32 - 2 = 30$.

Usable host per Subnet = 30.

Step 3: List the $/27$ Subnets

Subnet	Network Address	Host Range	Broadcast Address
1.	172.16.10.0	172.16.10.1 - 172.16.10.30	172.16.10.31
2.	172.16.10.32	172.16.10.33 - 172.16.10.62	172.16.10.63
3.	172.16.10.64	172.16.10.65 - 172.16.10.94	172.16.10.95
4.	172.16.10.96	172.16.10.97 - 172.16.10.126	172.16.10.127
5.	172.16.10.128	172.16.10.129 - 172.16.10.158	172.16.10.159
6.	172.16.10.160	172.16.10.161 - 172.16.10.190	172.16.10.191
7.	172.16.10.192	172.16.10.193 - 172.16.10.222	172.16.10.223
8.	172.16.10.224	172.16.10.225 - 172.16.10.254	172.16.10.255

Step 4: Determine the Subnet for IP Address 172.16.10.78

The IP Address 172.16.10.78 lies between 172.16.10.64 & 172.16.10.95

- Subnet: 172.16.10.64/27
- Valid Host range: 172.16.10.65 - 172.16.10.94
- Broadcast Address: 172.16.10.95

Result: The IP Address 172.16.10.78 belongs to the Subnet 172.16.10.64/27

Yes the 172.16.10.95 is within the same Subnet but it is the broadcast address not a usable host address.

Given IPv6 Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

compress the IPv6 Address

IPv6 Compression Rules:

Remove leading zeros in each 16-bit block

Replace one consecutive sequence of all-zero blocks with :: (only one)

Compressed Address: 2001:db8::ff00:42:8329

2. Expand the Compressed Address:

Expand :: back to the required number of 0000 blocks and restore leading zeros

Expanded Address: 2001:0db8:0000:0000:0000:ff00:0042:8329

3. Determine the Network prefix (/64)

A /64 prefix means

- First 64 bits (first four blocks) represent the Network
- Last 64 bits represent the Interface (Host) Identifier

Network Prefix:

2001:0db8:0000:0000/64

4. Calculate the Number of Host Address

Prefix length = /64

Remaining host bits = $128 - 64 = 64$ bits

Number of possible host addresses = 2^{64}

= 18,446,744,073,709,551,616